CH 4 — Power Series

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Power Series

🐳 **Info — Power Series Definition**

A **power series centered at $x = a$** is an infinite series of the form

$$\sum_{n=0}^{\infty} c_n(x-a)^n = c_0 + c_1(x-a) + c_2(x-a)^2 + \ldots$$

where x is a variable, and $a, c_n (n \in \mathbb{N} \cup 0)$ are constants. The set of all x values for which $\sum_{n=0}^{\infty} c_n(x-a)^n$ converges is called the **interval of convergence** for $\sum_{n=0}^{\infty} c_n(x-a)^n$

🐳 **Info — Power Series Properties**

Given any power series $\sum_{n=0}^{\infty} c_n(x-a)^n$, there are only 3 possibilities:

1. The power series converges only when $x = a$.
2. The power series converges absolutely $\forall x \in \mathbb{R}$.
3. There exists a strictly positive real number **radius of convergence** R s.t. the power series converges absolutely when $|x-a| < R$, diverges when $|x-a| > R$, and may converge or diverge(inconclusive) when $|x-a| = R$.

🐳 **Info — Interval of Convergence**

Given a power series $\sum_{n=0}^{\infty} c_n(x-a)^n$

1. The interval of convergence of the power series, denoted I is

$$I = \left\{ x_0 \in \mathbb{R} : \sum_{n=0}^{\infty} c_n(x-a)^n \text{ converges} \right\}$$

2. The radius of convergece denoted R
   - If $I = \{a\}, R = 0$
   - If $I = (-\infty, \infty), R = \infty$
   - Otherwise, R is the unique positive real number described in case 3 of the previous properties.

Examples:

Suppose $\sum_{n=0}^{\infty} c_n(x-2)^n$ diverges when $x = -1$ and converges $x = 4$. Determine if the following statements are must be true, must be falose or might be true

- The power series diverges when $x = 5$: Might be true

- The power series diverges when $x = 6$: Must be true
- The series $\sum_{n=0}^{\infty} c_n$ converges : Must be true
- The series $\sum_{n=0}^{\infty} (-1)^n 2^n c_n$ converges : Might be true

Representing Functions as Power Series

🐳 **Info — Abel's Theorem**

Let I denote the interval of convergence for the power series function $f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n$.

Then f is contiuous on I

Properties

Let $f(x) = \sum_{n=0}^{\infty} c_{n(x-a)}^{n}$ and $g(x) = \sum_{n=0}^{\infty} d_n (x-a)^n$

Let R_f and R_g be the radii of convergence for f, g respectively. Let I_f, I_g be the interval of convergence for f, g respectively. Then the following statements hold.

1. Let $h(x) = f(x) \pm g(x) = \sum_{n=0}^{\infty} (c_n \pm c_n)(x-a)^n$
   - If $R_f \neq R_g$, then the radius of convergence for $h(x) = \min\{R_f, R_g\}$ and the interval of convergence for $h(x) = I_f \cap I_g$.

   If $R_f = R_g$, then the radius of convergence for $h(x) = R_f$
2. Let $p(x) = (x-a)^k f(x) = \sum_{n=0}^{\infty} c_n (x-a)^{n+k}, k \in \mathbb{Z} \wedge k > 0$

   Then the radius of convergence for $p(x) = R_f$ and the interval of convergence for $p(x) = I_f$
3. Suppose that $a = 0$, i.e., $f(x) = \sum_{n=0}^{\infty} c_n x^n$ is a power series centred at $x = 0$

   $q(x) = f(bx^k) = \sum_{n=0}^{\infty} c_n b^n x^{kn}, b \in \mathbb{R}$

   Let $\{0\}, k \in \mathbb{R}^+$

   Then the radius of convergence for $q(x)$ is $R\left(\frac{R_f}{|b|}^{\frac{1}{k}}\right)$

   The interval of convergence for $q(x)$ is $\{x \in \mathbb{R} \mid bx^k \in I_f\}$

Examples:

Given $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ with $I = (-1, 1), R = 1$

1. $f(x) = \frac{1}{2-x} = \frac{1}{2}\left(\frac{1}{1-\frac{x}{2}}\right)$, center at $x = 0$

   $\frac{1}{2}\left(\frac{1}{1-\frac{x}{2}}\right) = \frac{1}{2}\sum_{n=0}^{\infty} \left(\frac{x}{2}\right)^n = \sum_{n=0}^{\infty} \frac{1}{2^{n+1}} x^n$

   $I = (-2, 2), R = 2$
2. $f(x) = \frac{x^2}{x+3} = \frac{x^2}{3}\left(\frac{1}{1-(-\frac{x}{3})}\right)$, center at $x = 0$

   $\frac{x^2}{3}\left(\frac{1}{1-(-\frac{x}{3})}\right) = \sum_{n=0}^{\infty} \left(-\frac{x}{3}\right)^n \frac{x^2}{3} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{n+2}}{3^{n+1}}$

   $I = (-3, 3), R = 3$

Differentiation and Integration on Power Series

🐳 **Info — Differentiation and Integration on Power Series**

Let $f(x) = \sum_{n=0}^{\infty} c_n (x-a)^n$ be a power series function with the radius of convergence $R > 0$

Then $f(x)$ is differentiable on the interval $(a-R, a+R)$

Moreover, the following statements hold.

1. $$f'(x) = \sum_{n=1}^{\infty} c_n n (x-a)^{n-1}, \qquad \forall x \in (a-R, a+R)$$

   Note: Interval of convergence might change when $R < \infty$ at the endpoints.

2. $$\int f(x)\,\mathrm{d}x = C + \sum_{n=0}^{\infty} c_n \frac{(x-a)^{n+1}}{n+1}, \qquad \forall x \in (a-R, a+R), C \in \mathbb{R}$$

   Note: Interval of convergence might change when $R < \infty$ at the enpoints.

3. The above power series for $f'(x)$ and $\int f(x)\,\mathrm{d}x$ both have the same radius of convergence of R

4. For any subinterval $[b, d] \subset (a-R, a+R)$, we have

$$\int_b^d f(x)\,\mathrm{d}x = \sum_{n=0}^{\infty} c_n \int_b^d (x-a)^n\,\mathrm{d}x = \sum_{n=0}^{\infty} c_n \frac{(x-a)^{n+1}}{n+1}\bigg|_b^d$$

All four statements continue to hold when the radius of convergence is $R = \infty$, in which we replace the interval with $\mathbb{R}$

Examples:

Given $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$ with $I = (-1, 1), R = 1$

1. $f(x) = \ln(x+1)$ centered at $x = 0$

   $\frac{\mathrm{d}}{\mathrm{d}x} \ln(x+1) = \frac{1}{1+x} = \frac{1}{1-(-x)}$

   $\frac{1}{1-(-x)} = \sum_{n=0}^{\infty} (-1)^n x^n$

   $\int \frac{1}{1+x}\,\mathrm{d}x = \int \sum_{n=0}^{\infty} (-1)^n x^n\,\mathrm{d}x = \sum_{n=0}^{\infty} \int (-1)^n x^n\,\mathrm{d}x = C + \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1}$

   $\ln(x+1) = \sum_{n=0}^{\infty} (-1)^n \frac{x^{n+1}}{n+1}$

   $R = 1, I = (-1, 1]$

2. $f(x) = \frac{x}{(1-2x)^2}$

   $\frac{\mathrm{d}}{\mathrm{d}x} \frac{1}{1-x} = \frac{1}{(1-x)^2}$

   $\frac{1}{(1-2x)^2} = \sum_{n=0}^{\infty} n 2^{n-1} x^{n-1}, \quad |2x| < 1 \implies |x| < \frac{1}{2}$

   $\frac{x}{(1-2x)^2} = \sum_{n=0}^{\infty} n 2^{n-1} x^n$

   $R = 1$

3. Consider $f(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}, x \in \mathbb{R}$, and this function is the IVP $y' = y, y(0) = 1$

   $f'(x) = \sum_{n=1}^{\infty} \frac{nx^{n-1}}{(n-1)!} = \sum_{n=0}^{\infty} \frac{x^n}{n!}$

   $f(0) = 1$

   $f(x) = e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}, x \in \mathbb{R}$

Taylor Series

🐳 **Info — Taylor Series**

Uniqueness of Power Series Suppose that $f(x) = \sum_{n=0}^{\infty} c_n(x-a)^n, \forall x \in (a-R, a+R), R > 0$.

Then for every $n > 0$, we have $c_n = \frac{f^n(a)}{n!}$

Taylor Series

Let $f(x)$ be a m degree polynomial function

Its Taylor Series centered at $x = a$ is

$$f(x) = T_{m,a}(x) = \sum_{n=0}^{m} \frac{f^n(a)(x-a)^n}{n!}$$

- For every integer $j \in \{0, 1, 2, 3, ..., m\}, T_{m,a}^{j}(a) = f^j(a)$

Maclaurin Series

$$f(x) = \sum_{n=0}^{\infty} \frac{f^n(0)x^n}{n!}$$

is a Maclaurin series for $f(x)$

Examples:

1. $f(x) = \sin x, a = 0$

   $f(0) = 0, f'(0) = 1, f''(0) = 0, f'''(0) = -1, f^4 = 0$

   $\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \quad \forall x \in \mathbb{R}$

   $\lim_{n\to\infty} \left| \frac{(-1)^{n+1} x^{2n+3}}{(2n+3)!} \cdot \frac{(2n+1)!}{(-1)^n x^{2n+1}} \right| = \lim_{n\to\infty} \frac{|x|^2}{2n+2}(2n+3) = |x|^2 \cdot 0 = 0 < 1.$

   Converges by Ratio Test.

   The power series converges for all $x \in \mathbb{R}$

2. $f(x) = \cos(x), a = 0$

   Since $\frac{\mathrm{d}}{\mathrm{d}x} \sin x = \cos x$.

   $\cos(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}, \quad \forall x \in \mathbb{R}$

💡 **Tip — Common Series**

$$\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$$

$$\cos(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}$$

$$e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

🐳 **Info — Lagrange's Remainder Formula**

Let m be a nonnegative integer. Suppose that f^{m+1} is continuous on an open interval I that contains a. If $x \in I$, then there is a real number c between a and $x(i.e., c \in [a, x], x \geq a \vee c \in [x, a], x < a)$ s.t.

$$f(x) - T_{m,a}(x) = \frac{f^{m+1}(c)}{(m+1)!}(x-a)^{m+1}$$

Example:

$f(x) = \sin(x), a = 0$

Proof

Let $x \in \mathbb{R}, m \in \mathbb{N}$.

Since $f^{m+1}(x)$ is one of $\pm\cos(x), \pm\sin(x)$, $f^{m+1}(x)$ is continuous on $\mathbb{R}$

By LRF, $\exists c \in [0, x]$ s.t. $\sin(x) - T_{m,0}(x) = \frac{f^{m+1}(c)}{(m+1)!}x^{m+1}$.

Thus $|f^{m+1}(c)| \leq 1 \implies |\sin(x) - T_{m,0}(x)| = |\frac{f^{m+1}(c)}{(m+1)!}x^{m+1}| \leq \frac{|x|^{m+1}}{(m+1)!}$.

Since $\lim_{n\to\infty} \frac{|x|^{m+1}}{(m+1)!} = 0$, by Squeeze Theorem, $\lim_{n\to\infty} |\sin(x) - T_{m,0}(x)| = 0$.

Therefore $f(x) = \lim_{n\to\infty} T_{m,0} = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$

$\square$

🐳 **Info — Taylor's Inequality**

Let m be a nonnegative integer. Suppose that f^{m+1} is continuous on an open interval I that contains a.

If $|f^{m+1}| \leq K$ for every $x \in I$, then for every $x \in I$,

$$|f(x) - T_{m,a}(x)| \leq \frac{K|x-a|^{m+1}}{(m+1)!}$$

Examples:

1. Let $f(x) = \sqrt{1+x}$

$f(x) = (1+x)^{\frac{1}{2}} \implies f(0) = 1$

$f'(x) = \frac{1}{2}(1+x)^{-\frac{1}{2}} \implies f'(0) = \frac{1}{2}$

$f''(x) = -\frac{1}{4}(1+x)^{-\frac{3}{2}} \implies f''(0) = -\frac{1}{4}$

$f'''(x) = \frac{3}{8}(1+x)^{-\frac{5}{2}}$

$T_{2,0}(x) = 1 + \frac{1}{2}x - \frac{1}{4 \cdot 2!}x^2$

$T_{2,0}(1) = \frac{11}{8}$

$|f^3(c)| \leq |f^3(0)| = \frac{3}{8}$

Therefore $|f(1) - T_{2,0}(1)| \leq \frac{\frac{3}{8}}{6} = \frac{1}{16}$

2. Find Taylor Polynomial for $f(x) = x^2 + x + 1$ centered at $x = 1$

   $3 + 3(x-1) + (x-1)^2$

3. Suppose that $T_{3,-1}(x)$ for a function f is given by $T_{3,-1} = 2 - 3(x+1)^2 + \frac{1}{5}(x+1)^3$.

   Find $f'(-1), f''(-1)$

   $f'(-1) = 0$

   $f''(-1) = -3 * 3! = -6$

4. $f(x) = x^3 e^{x^2}, f^{63}(0)$

   $e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!} \implies x^3 e^{x^2} = \sum_{n=0}^{\infty} \frac{x^{2n+3}}{n!}$, which is the Maclaurin Series for the given function which is unique.

   $f'(63)\frac{0}{63!} \implies \frac{1}{30!} \implies \frac{30!}{63!}$

5. Let $f(x) = x^{\frac{2}{3}}$. Determine $T_{2,8}(x)$ to determine an upper bound on the error in using $T_{2,8}$ to estimate $f(x), x \in (5, 11)$

   Let $f(x) = x^{\frac{2}{3}}, f'(x) = \frac{2}{3}x^{-\frac{1}{3}}, f''(x) = -\frac{2}{9}x^{-\frac{4}{3}}, f'''(x) = \frac{8}{27}(x)^{-\frac{7}{3}}$

   $T_{2,8} = 4 + \frac{1}{3}(x-8) - \frac{1}{144}(x-8)^2$

   Note that $f'''(x)$ is continuous on $I = (5, 11)$ containing 8.

   $|f^3(x)| = f^3(x) \leq f^3(5) = \frac{8}{27} \cdot \frac{1}{5^{\frac{7}{3}}}$

   Therefore, for all $x \in I$, $|f(x) - T_{2,8}(x)| \leq \frac{\frac{8}{27} \cdot \frac{1}{5^{\frac{7}{3}}}}{3!} |x-8|^3 < \frac{\frac{8}{27} \cdot \frac{1}{5^{\frac{7}{3}}}}{3!} 3^3 = \frac{4}{3 \cdot 5^{\frac{7}{3}}}$

Binomial Series

🐳 **Info — Binomial Coefficient**

Let α be a real number and let n be some positive integers

$$\binom{\alpha}{n} = \frac{(\alpha)(\alpha-1)(\alpha-2)...(\alpha-n+1)}{n!} = \frac{\alpha!}{n!(\alpha-n)!}$$

Note:

$\binom{\alpha}{0} = 1, \binom{\alpha}{-n} = 0$

$\binom{\alpha}{n+1}(n+1) + \binom{a}{n}n = \binom{\alpha}{n}\alpha$

🐳 **Info — Generalized Binomial Theorem**

Let α be a real number. If $|x| < 1$, then $(1+x)^\alpha = \sum_{n=0}^{\infty}\binom{\alpha}{n}x^n = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2}x^2 + ...$

If $|x| > 1$ and α is a nonnegative integer, then the series diverges

Examples:

1. Use the generalized binomial theorem to derive the power series representation of $f(x) = \frac{1}{1-x}$ centered at $x = 1$

   Note that using the binomial theorem for $|x| < 1, (1+x)^{-1} = \sum_{n=0}^{\infty}\binom{-1}{n}x^n$

   $\implies \frac{1}{1-x} = \sum_{n=0}^{\infty}\binom{-1}{n}(-1)^n x^n = \sum_{n=0}^{\infty}\frac{(-1)(-2)...(-n)}{n!}(-1)^n x^n = \sum_{n=0}^{\infty} x^n$

2. Determine Maclaurin series for $f(x) = \sqrt{1+x}$

   $(1+x)^{\frac{1}{2}} = \sum_{n=0}^{\infty}\binom{\frac{1}{2}}{n}x^n$

   $$= 1 + \sum_{n=1}^{\infty}\frac{(\frac{1}{2})(\frac{1}{2}-1)...(\frac{1}{2}-(n-1))}{n!}x^n = 1 + \frac{1}{2} + \sum_{n=2}^{\infty}\frac{(-1)^{n-1}(2n-3)}{2^n}\frac{x^n}{n!}$$

   We often write the product $(2n-3) \times (2n-5) \times ...3 \times 1 = (2n-3)!!$ meaning that all the terms in the entire factorial have the same parity (even or odd) as the given term

3. Compute $f^6(0), f^7(0)$ for $f(x) = x\sqrt{1+x^3}$

   $x(1+x^3)^{\frac{1}{2}} = \sum_{n=0}^{\infty}\binom{\frac{1}{2}}{n}x^{3n+1}$

   $3n+1 = 6, n \notin \mathbb{N} \implies f^6(0) = 0$

   $\frac{f^7(0)}{n!} \implies n = 2 \implies \binom{\frac{1}{2}}{2} = \frac{-\frac{1}{4}}{2!} = -\frac{1}{8} \implies f^7(0) = -\frac{1}{4}$

Applications of Taylor Series

Examples:

1. Finding sum of given series

   - $\sum_{n=1}^{\infty}\frac{2^n}{n!} = \sum_{n=0}^{\infty}\frac{2^n}{n!} - 1 = e^2 - 1$
   - $\sum_{n=0}^{\infty}\frac{(-1)^n\pi^{2n}}{4^n(2n)!} = \sum_{n=0}^{\infty}\frac{(-1)^n}{(2n)!}\left(\frac{\pi}{2}\right)^{2n} = \cos\left(\frac{\pi}{2}\right) = 0$
   - $\sum_{n=0}^{\infty}\frac{1}{n3^n} = \sum_{n=1}^{\infty}\frac{(-1)^{2n-1}}{n}\left(\frac{1}{3}\right)^n = \sum_{n=1}^{\infty}\frac{-1}{n}\left(\frac{1}{3}\right)^n = -\ln\left(\frac{2}{3}\right) = \ln 3 - \ln 2$
   - $\sum_{n=0}^{\infty}\frac{n+1}{n!}x^n, \forall x \in \mathbb{R}$

$\sum_{n=0}^{\infty} \frac{n}{n!} x^n + \sum_{n=0}^{\infty} \frac{1}{n!} x^n = \sum_{n=1}^{\infty} \frac{n}{n!} x^n + e^x = \sum_{n=0}^{\infty} \frac{x^{n+1}}{n!} = xe^x + e^x$

2. Express $\int \sin(x^2)\,\mathrm{d}x$

   $\sin(x) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}$

   $\sin(x^2) = \sum_{n=0}^{\infty} \frac{(-1)^n x^{4n+2}}{(2n+1)!}$

   $\int \sin(x^2)\,\mathrm{d}x = C + \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} \frac{x^{4n+3}}{4n+3} = C + \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)! \cdot 4n+3} x^{4n+3}$

3. Express numerical series of $\int_0^1 \sin(x^2)\,\mathrm{d}x$

   $\int_0^1 \sin(x^2)\,\mathrm{d}x = \sum_{n=0}^{\infty} \left[\int_0^1 \frac{(-1)^n x^{4n+2}}{(2n+1)!}\right] = \sum_{n=0}^{\infty} \left(\frac{(-1)^n x^{4n+3}}{(2n+1)! \cdot (4n+3)}\right)\Big|_0^1 = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)! \cdot (4n+3)}$

4. Use power series to evaluate $\lim_{x\to 0} \frac{e^x - 1}{x}$

   $\lim_{x\to 0} \frac{\sum_{n=0}^{\infty} \frac{x^n}{n!} - 1}{x} = \lim_{x\to 0} \frac{\sum_{n=0}^{\infty} \frac{x^n}{n!} - 1}{x} = \lim_{x\to 0} \left(\sum_{n=0}^{\infty} \frac{x^{n-1}}{n!}\right) = 1$

Big-O Notation

🐳 **Info — Big-O Notation**

Let a denote a real number or ∞. Assume that $f(x)$ and $g(x)$ are two functions that are defined for all real x values that are near a but are not necessarily defined when $x = a$. More precisely, if $a \in \mathbb{R}$, we assume that $f(x)$ and $g(x)$ are both defined for all x in the union

$$I = (a - \delta, a) \cup (a, a + \delta) = \{x \in \mathbb{R} \mid 0 < |x - a| < \delta\},$$

where δ is some strictly positive real number. If $a = \infty$, we assume that $f(x)$ and $g(x)$ are both defined for all x in some open interval $I = (c, \infty)$, where $c \in \mathbb{R}$ denoted

$$f(x) = O(g(x)) \text{ as } x \to a$$

if there exists a real constant $M > 0$ s.t.

$$|f(x)| \leq M|g(x)|$$

for every $x \in I$.

Examples:

1. $x = O(x^2)$ as $x \to \infty$.

   Let $M = 1$ and $D = (1, \infty)$. Then $|x| = 1|x| < |x||x| = |x|^2 = 1|x^2|, x \in D$

2. $x^2 = O(x)$ as $x \to 0$.

   Let $M = 1$ and $D = (-1, 0) \cup (0, 1)$. Then $|x^2| = |x|^2 = |x||x| < 1|x|, x \in D$

🐳 **Info — Theorems**

1. Let a denote a real number or ∞. If $f(x) = O(g(x))$ as $x \to a$ and $g(x) = O(h(x))$ as $x \to a$, then $f(x) = O(h(x))$ as $x \to a$
2. Let a, b be real numbers or ∞. Assume that $f(x) = O(g(x))$ as $x \to a$, and $\lim_{x \to b} h(x) = a$. If a is a real number, further assume that $h(x) \neq a$ for every x sufficiently close to b. (More precisely, if $b \in \mathbb{R}$, assume that $h(x) \neq a$ for every x satisfying $0 < |x - b| < \delta$ for some constant $\delta > 0$. If $b = \infty$, assume $h(x) \neq a$ for every x satisfying $x > c \in \mathbb{R}$). Then $f(h(x)) = O(g(h(x)))$ as $x \to b$

🐳 **Info — Properties**

🐳 **Info — Taylor's Inequality, Big-O**